

UBON RATCHATHANI, Thailand

WMO: 484070

Lat: 15.251N

Lon: 104.870E

Elev: 124

StdP: 99.85

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 15.3	(c) 16.8	(d) 10.2	(e) 7.8	(f) 20.1	(g) 11.8	(h) 8.7	(i) 20.6	(j) 9.4	(k) 24.4	(l) 8.5	(m) 24.1	(n) 2.8	(o) 20	(p) 0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.4	37.4	25.8	36.4	25.6	35.5	25.4	27.6	32.7	27.2	32.2	26.9	31.7	2.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	22.0	28.8	26.1	21.8	28.8	25.9	21.5	28.7	88.5	33.1	86.9	32.5	85.5	32.0	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.0	6.0	5.1		13.0	39.0	1.4	1.0	12.0	39.8	11.1	40.4	10.3	41.0	9.3	41.7	
			WB	11.4	28.7	1.5	0.8	10.3	29.2	9.4	29.7	8.5	30.2	7.4	30.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	24.9	26.8	29.1	30.4	29.7	29.1	28.4	28.2	27.9	27.6	26.5	24.8
	DBStd	2.52	2.28	2.66	2.39	1.97	1.62	1.34	1.32	1.22	1.25	1.26	1.91	2.36
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6487	461	471	591	611	611	573	571	564	537	546	494	458
	CDD18.3	3445	203	237	332	361	352	323	312	306	287	287	244	200
	CDH23.3	38223	1977	2667	4168	4728	4274	3791	3433	3302	2949	2873	2287	1774
	CDH26.7	17164	856	1322	2239	2608	2082	1672	1353	1224	1032	1112	952	715
Wind	WSAvg	2.0	2.2	1.9	1.8	1.6	1.6	1.8	2.0	2.0	1.6	1.9	2.6	2.8
Precipitation	PrecAvg	1644	1	10	43	78	236	255	305	329	271	86	27	3
	PrecMax	2239	6	58	181	183	644	480	465	572	499	268	82	17
	PrecMin	1294	0	0	0	11	70	131	120	128	169	4	0	0
	PrecStd	262	2	17	55	49	144	83	113	107	91	66	26	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.6	36.6	38.1	39.2	38.0	36.1	35.0	34.1	34.0	33.9	34.8	34.1
		MCWB	23.5	24.5	25.4	26.1	26.2	26.3	26.1	26.7	26.6	25.6	25.3	24.3
	2%	DB	33.5	35.3	37.0	37.9	36.4	34.8	33.8	33.1	32.9	33.1	33.6	33.0
		MCWB	23.2	24.1	25.1	26.0	26.3	26.4	26.2	26.4	26.2	25.3	24.8	23.7
	5%	DB	32.4	34.4	36.0	36.8	35.2	33.8	32.9	32.2	32.1	32.2	32.7	31.9
		MCWB	22.6	23.9	24.9	26.0	26.2	26.3	26.0	26.1	26.0	25.1	24.4	23.2
	10%	DB	31.2	33.2	35.0	35.5	34.0	32.9	31.9	31.5	31.2	31.6	31.5	30.7
		MCWB	22.0	23.5	24.7	25.8	26.1	26.2	25.8	26.0	25.9	25.0	24.0	22.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	26.4	27.2	28.1	27.7	27.7	27.5	27.7	27.6	27.0	26.6	25.5
		MCDB	31.8	33.1	34.8	35.7	33.9	32.3	31.4	31.3	31.1	31.1	31.6	31.9
	2%	WB	24.5	25.7	26.4	27.2	27.2	27.2	26.9	27.1	27.0	26.5	26.0	24.7
		MCDB	30.8	32.8	34.0	34.7	33.1	31.9	31.1	30.8	30.6	30.5	31.1	31.0
	5%	WB	23.8	25.1	25.9	26.7	26.8	26.8	26.6	26.7	26.6	26.1	25.5	24.1
		MCDB	30.1	32.1	33.6	34.0	32.4	31.4	30.6	30.3	30.1	30.1	30.4	30.1
	10%	WB	23.1	24.5	25.5	26.3	26.5	26.5	26.2	26.4	26.2	25.7	25.0	23.3
		MCDB	29.5	31.0	32.7	33.2	31.8	31.0	30.2	29.9	29.6	29.6	29.6	29.3
Mean Daily Temperature Range	5% DB	MDBR	12.9	12.8	11.8	10.4	8.8	7.7	7.0	6.6	6.8	8.3	10.0	11.4
		MCDBR	13.4	13.0	12.4	11.5	10.3	8.9	8.2	7.6	8.0	9.3	11.0	12.2
		MCWBR	4.7	4.2	3.9	3.2	2.9	2.8	2.7	2.7	2.8	3.3	3.8	4.3
	5% WB	MCDBR	12.3	11.4	11.3	10.6	9.1	8.2	7.5	7.1	7.3	8.2	9.9	11.4
		MCWBR	4.7	4.3	3.9	3.4	3.1	2.9	2.9	2.9	2.9	3.2	3.7	4.4
Clear-Sky Solar Irradiance	taub		0.463	0.587	0.665	0.611	0.500	0.466	0.462	0.462	0.494	0.505	0.452	0.443
	taud		2.150	1.818	1.685	1.851	2.189	2.299	2.309	2.302	2.201	2.145	2.277	2.247
	Ebn at Noon		826	739	695	735	808	828	833	842	818	799	832	834
	Edh at Noon		147	214	251	213	149	132	131	134	148	152	129	131
All-Sky Solar Radiation	RadAvg		5.15	5.42	5.33	5.58	5.50	5.10	4.79	4.60	4.59	5.03	5.00	4.90
	RadStd		0.25	0.25	0.18	0.25	0.28	0.27	0.41	0.32	0.43	0.36	0.20	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	+0.29	N/A	N/A	N/A	+0.22	+0.20	N/A	N/A	+103	+102

Nomenclature: See separate page